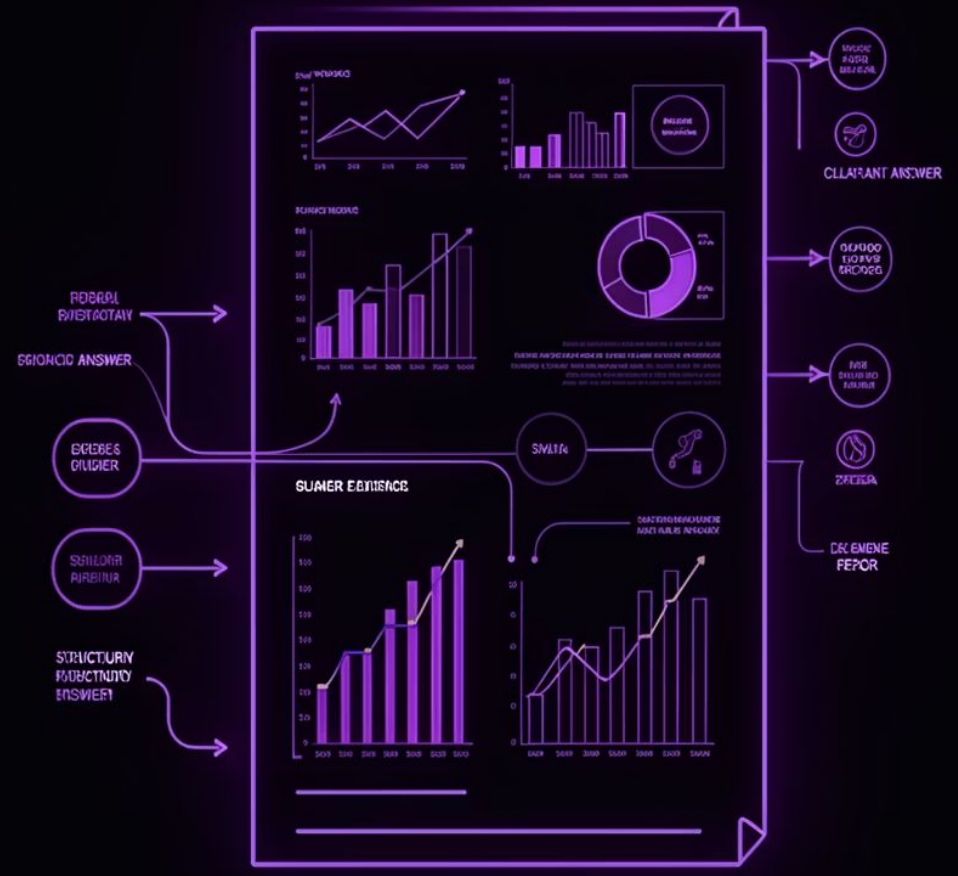
ENTERPRISE SUSTAINABILITY · 2026

Enterprise Sustainability RAG

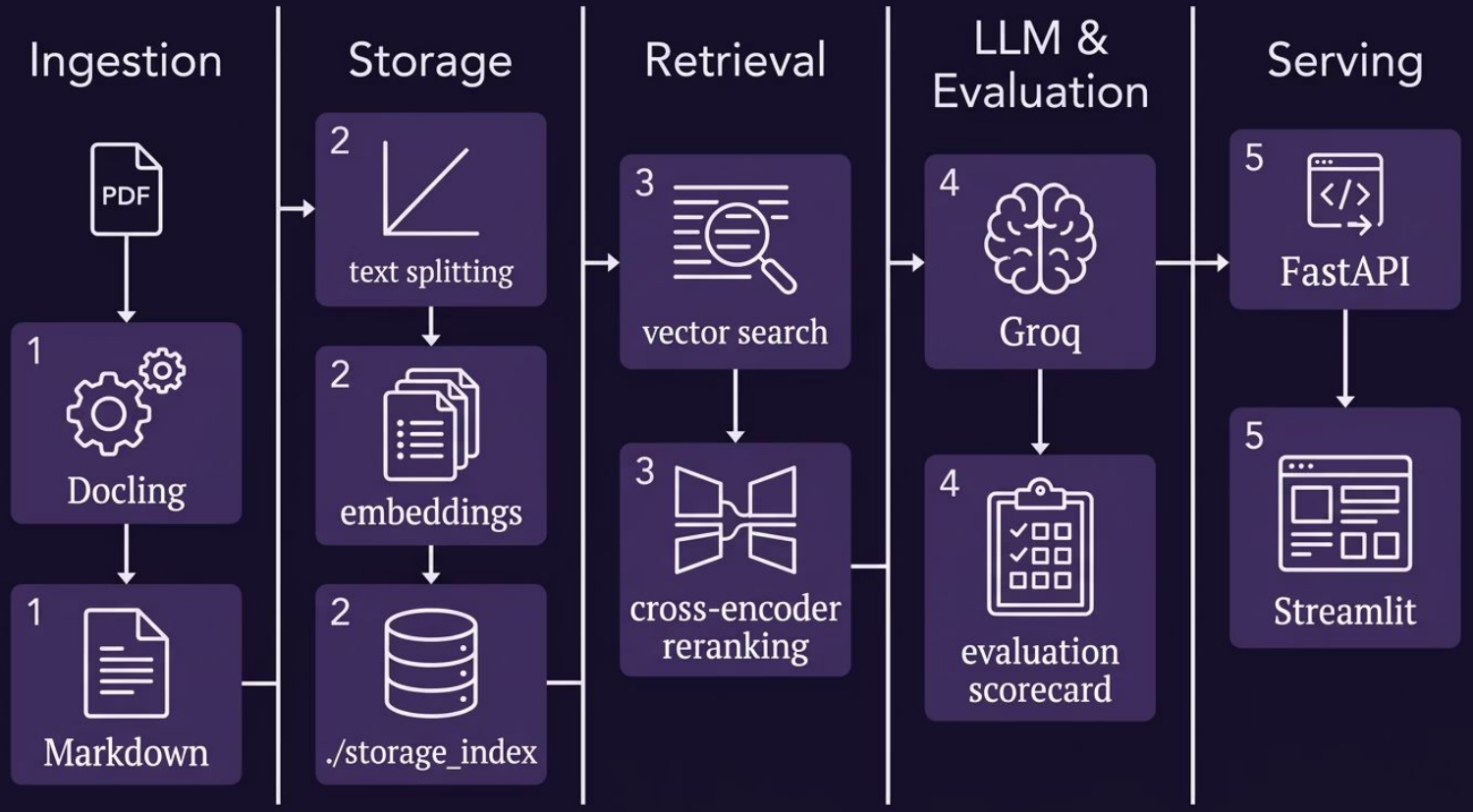
Trusted answers from complex ESG reports.



Made with GAMMA

From PDF to trusted answer

One horizontal path connects document structure, searchable evidence, evaluation, and the user interface.



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Docling keeps the document structure

The pipeline preserves the shape of complex ESG reports before search begins.



Input

Apple, Unilever, and Coca-Cola reports



Output

Markdown with tables, headers, and data blocks



Speed

cached .md files skip repeat work



Tool

PyPdfiumDocumentBackend

Made with GAMMA

Chunks become searchable evidence

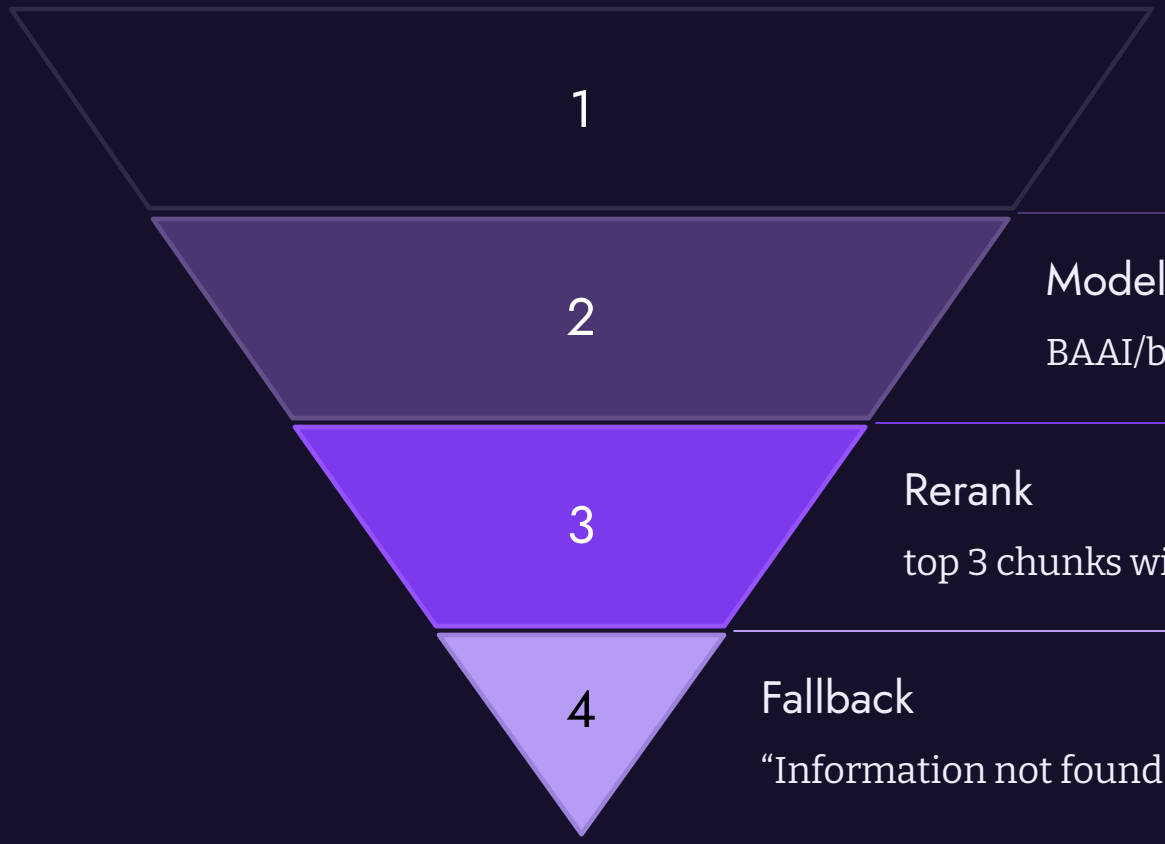
Each passage carries enough context to retrieve the right source later.

Field	Value	Why it matters	System location
Chunk size	1024	Readable evidence units	Vector index
Overlap	200	Context across boundaries	Chunking stage
Metadata	source file, chunk index, company	Traceable retrieval	Every chunk
Storage	./storage_index	Persistent evidence base	Disk persistence
Restore	under 1 second	Fast cache hits	Restore path

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Reranking cuts the noise

Broad search finds candidates. A cross-encoder keeps the three chunks that matter.



Search

15 chunks from VectorStoreIndex

Model

BAAI/bge-base-en-v1.5

Rerank

top 3 chunks with BAAI/bge-reranker-base

Fallback

“Information not found in context.”

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05 — CHECK

Three checks before release

Every answer is scored before it reaches the user.



Faithfulness

facts match the source; score 1–5; threshold ≥3



Relevance

answer matches the question

Conciseness

clear, brief, no repetition

Judges: LlamaIndex + Groq

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How does Unilever manage plastic waste?

Or type your custom analytical question here:

How does Unilever manage plastic waste?

Run Grounded RAG Pipeline Engine

Strategic Audit Response

Unilever's approach to managing plastic waste

Action	Description	Source
Reduce virgin plastic	Switch to recycled plastic, design lighter packaging, develop alternative materials and formats.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 194)
Introduce plastic-free alternatives	Launched laundry sheets and product capsules packaged in cardboard to eliminate or cut plastic use.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 194)
Reuse-refill pilots	Ran >50 pilots worldwide since 2021 to test reusable and refillable packaging models.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 194)
Industry collaboration	Participates in initiatives led by the World Economic Forum, Ellen MacArthur Foundation, and Consumer Goods Forum to scale reusable packaging.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 194)
Next-generation packaging	Develops reusable, recyclable or compostable packaging; separates rigid from flexible formats to apply appropriate solutions.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 195)
Recyclable pump launch	Introduced a recyclable pump for Vaseline bottles in North America (2024).	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 195)
Future Flexibles programme	Explores recyclable and compostable paper-based materials; moves some products (e.g., Knorr bouillon cubes in the UK) to paper-plastic solutions.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 195)
Support collection & processing	backs initiatives that expand waste-management systems, prevent plastic leakage and reduce micro-plastic release.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 195)
Policy advocacy	Supports Extended Producer Responsibility (EPR) schemes, packaging-design rules, recycled-content targets, and a legally binding UN treaty to harmonise global standards.	(Source File: unilever-annual-report-and-accounts-2024-243-293.md, chunk 770)

Summary

Unilever manages plastic waste by (1) minimizing virgin-plastic use through material substitution and lighter designs, (2) introducing plastic-free or reduced-plastic product formats, (3) piloting and scaling reusable/refill models, (4) developing recyclable, compostable and paper-based packaging, (5) supporting collection and (6) advocating for regulatory frameworks such as EPR schemes and a UN treaty to enforce circular-economy standards.

Capstone Step 7 Qualitative Evaluation Scorecard

Faithfulness Score

5/5

↑ PASSED

YES

Relevance Score

5/5

↑ PASSED

YES

Conciseness Score

4/5

↑ PASSED

The answer presents the information in a compact table and a brief summary without unnecessary format adds some length that could be slightly more concise.

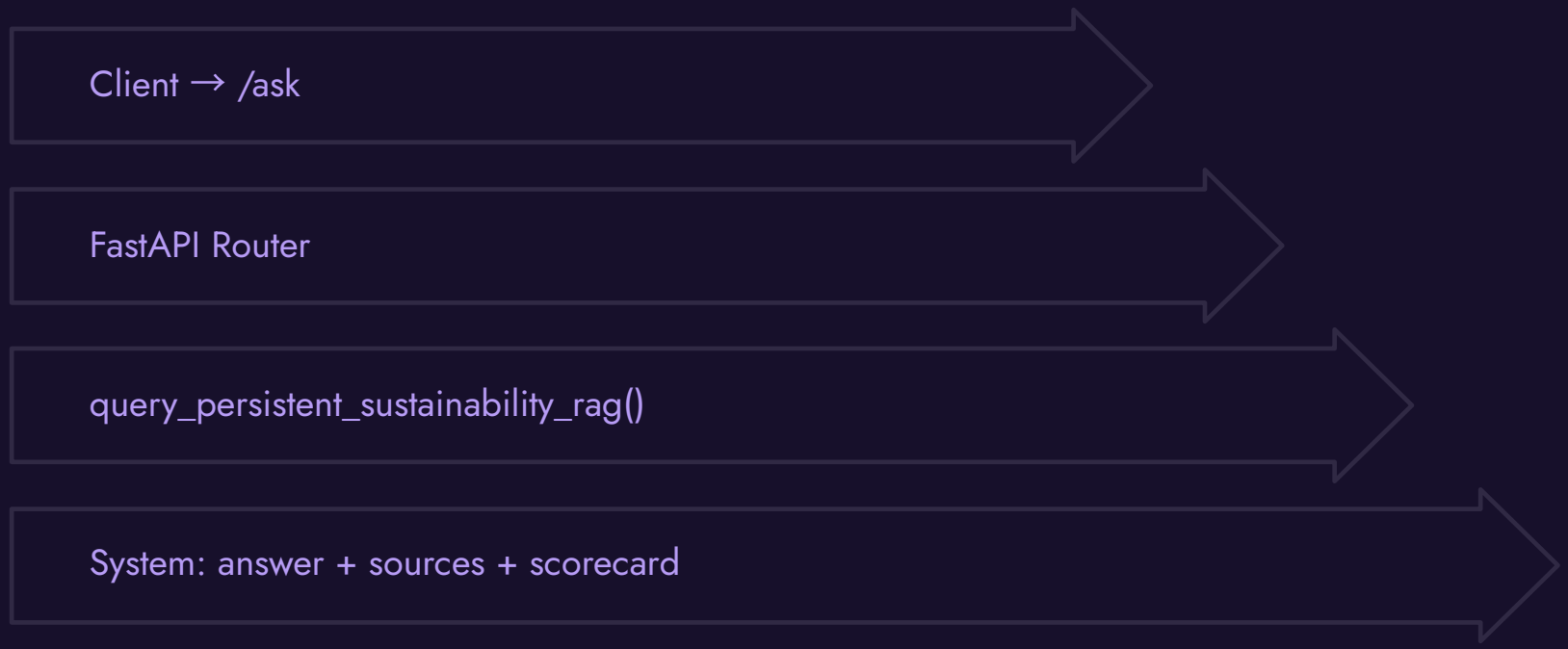
Supporting Evidence Blocks (Top Reranked Context Excerpts)

Excerpt 1 (Rerank Score: 0.9845) Excerpt 2 (Rerank Score: 0.9340) Excerpt 3 (Rerank Score: 0.9193)

** Document Source file:** `unilever-annual-report-and-accounts-2024-243-293.md`

Made with GAMMA

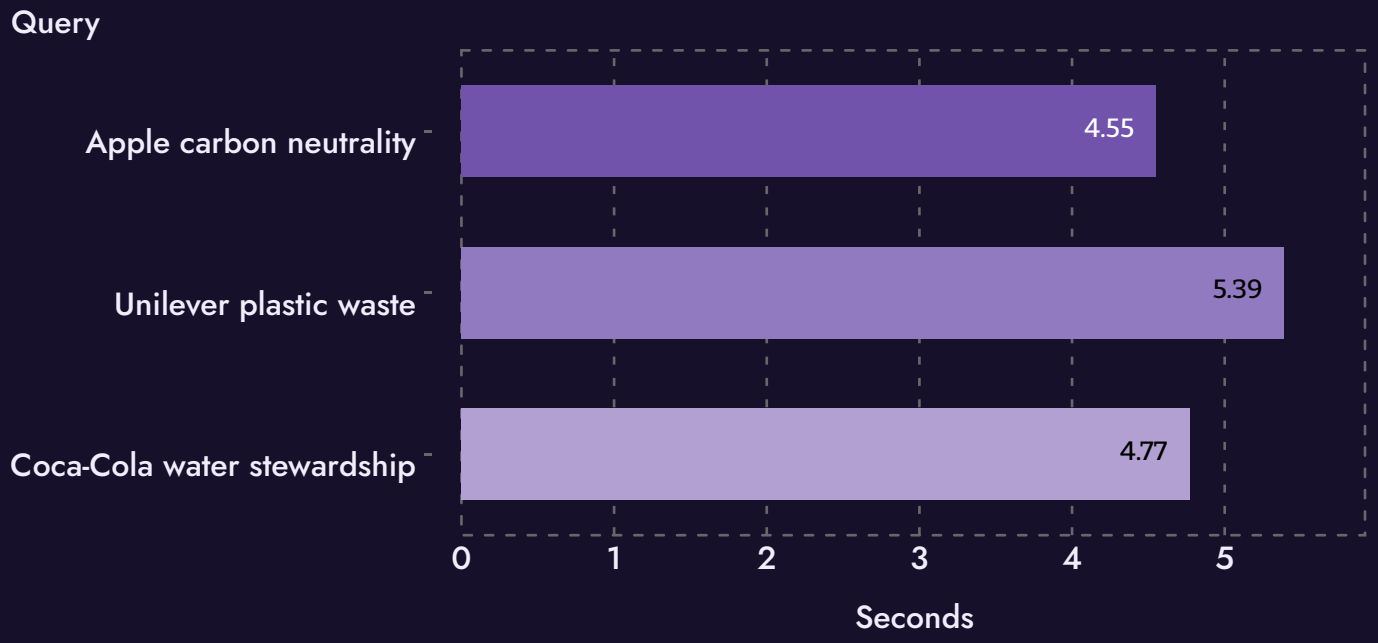
FastAPI connects the system to users



/ask	QueryPayload	answer, sources, scorecard	Streamlit
Endpoint	Input	Output	UI

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Every test query finished in under 6 seconds



Under 6 seconds

Apple: carbon neutrality — 4.55s

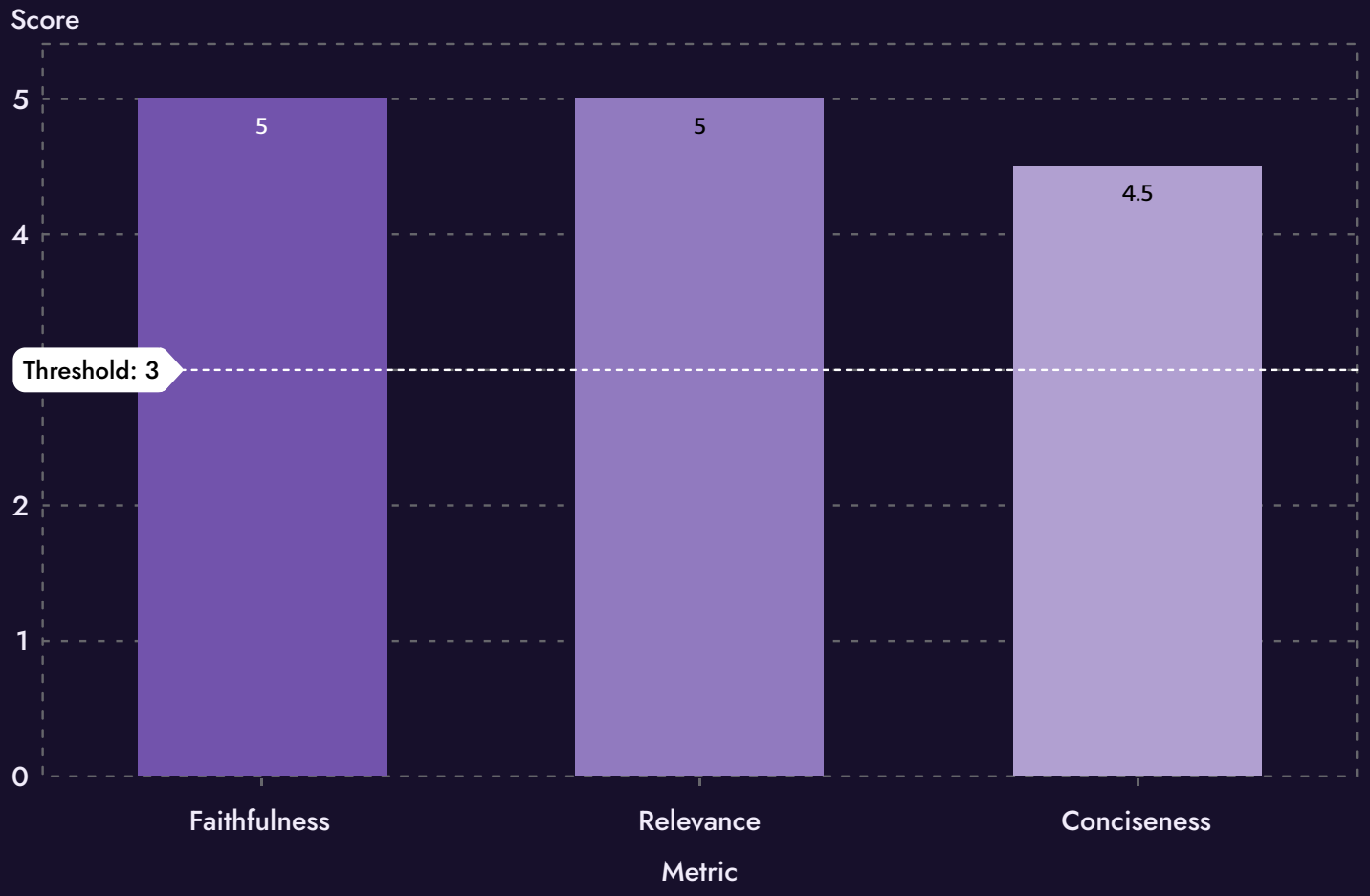
Unilever: plastic waste — 5.39s

Coca-Cola: water stewardship — 4.77s

Water metric: 1.78 L/L baseline

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Quality scores stay high



Passed the release bar

Faithfulness: 5/5 — passed

Relevance: 5/5 — passed

Conciseness: 4–5/5 — passed

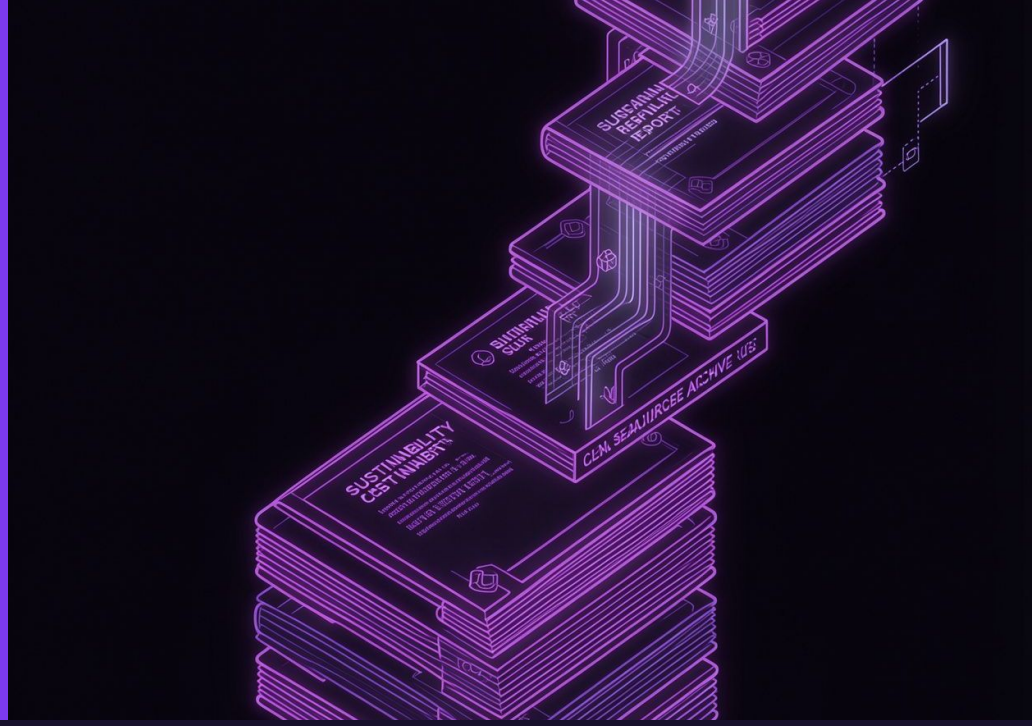
Result: no unsupported claims detected across test workloads

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09 - NEXT

Next: scale the evidence base

Q&A



Expand

multi-year report archives

Add

hybrid BM25 + vector search

Keep

disk persistence and source grounding

Cost, Quality and Latency

Optimize Token Cost, Response Latency and Quality

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